Assignment 2

The Project: "TechShare" Equipment Manager

Scenario: You have been hired to build a simple command-line interface (CLI) tool for the university lab to manage the borrowing of equipment (laptops, microscopes, sensors, etc.).

You must write a **single Python script** that performs the following four tasks.

Task 1: Inventory Management (2 Points)

Create a global list called inventory. Then, define a function called **add_gadget**.

- **Parameters:** The function must accept a mandatory argument name and arbitrary keyword arguments (**kwargs) for details (e.g., serial_num, brand).
- **Logic:** Inside the function, create a dictionary containing the name, a status set to "available", and all items from kwargs. Append this dictionary to the inventory list.

Task 2: Lending Logic & Error Handling (3 Points)

Define a function called **checkout_gadget(gadget_name).**

- **Logic:** Loop through the inventory. If the item is found and is "available", change its status to "borrowed" and print a success message.
- **Exceptions:**
  - If the item is found but is *already* borrowed, check logic should raise a ValueError.
  - If the item is not found at all, raise a KeyError (or return a sentinel value that causes an error).
- **Implementation:** In your main execution block (Task 4), you must call this function inside a try-except block to catch these specific errors and print user-friendly warnings.

Task 3: Reporting & File I/O (3 Points)

Define a function called generate_report(filename).

- **Statistics:** Calculate the total number of items, the count of "available" items, and the count of "borrowed" items.
- **List Comprehension:** Use a list comprehension to create a list of names of all currently borrowed items.
- **File Output:** Use the with open(...) context manager to write these statistics and the list of borrowed items into a text file.

Task 4: The Main Application Loop (2 Points)

Create the main execution block (not inside a function).

- Use a while True loop to display a menu:
  1. Add Equipment
  2. Checkout Equipment
  3. Generate Report
  4. Exit
- Capture user input and call the appropriate functions.
- Ensure the program exits cleanly when the user selects option 4.

Good luck! Start by sketching out your data structure on paper before you code.